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Your Name

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Abstract

1 Introduction

I had browsed through existing works in recent 3 years in the field of Intelligent Vehicle and summarized the topic and the trend of the research. [1–12]

There are several main problems in the field of Intelligent Vehicle, including bandwidth & communication bottlenecks, latency & temporal asynchrony, heterogeneity & cross-model compatibility, infrastructure-assisted efficiency (V2I), robustness, security & reliability, and sim-to-real gap & empirical deployment constraints.

Then I focus on the problem of latency & temporal asynchrony in cooperative perception. These two challenges are critical to spatial-temporal data consistency. Latency refers to the delay from sensation to fusion and decision-making, which can be caused by network transmission, data processing, and algorithmic computation. This latency comes from bottlenecks in communication, computation, and synchronization, which can lead to outdated or incomplete information being used for decision-making. Temporal asynchrony refers to the misalignment of data timestamps from different sources, including RSUs and CAVs, which can lead to inconsistencies in the perception results.

Usually, data are fused trivially by addition or concatenation, which can lead to spatiotemporal misalignment and inaccurate perception results. Moving objects are predicted to be in wrong historical positions instead of their current positions, leading to phantom detections, tracking fragmentation, and dangerous miscalculations in downstream motion planning.

And nowadays, there are several methods to address the latency and temporal asynchrony problem in cooperative perception, each compensating from different aspects. [13–18]

These include predicting current features using historical information, performing spatial warping of features via motion information or BEV flow, and leveraging trajectory information to establish correspondences across different timestamps. However, these methods align features in a single frame, which may not fully capture the temporal dynamics of moving objects and may not be robust to occlusions or missing data. Therefore, recently research moves toward object level.

2 Object level Alignment

Object-Level Alignment 的核心概念是利用物體本身作為不同 Agent 與不同 timestamp 之間的 correspondence reference，例如 object state、bounding box、track、trajectory 或 object-level anchor。相較於直接對整張 feature map 進行 temporal compensation，object-level 方法可以針對個別物體估計其位置、速度與時間變化，再將歷史物體資訊 propagation 至當前 timestamp。例如，CoAnchor 利用 object-level spatiotemporal anchors 建立跨 Agent 的 correspondence，先進行 temporal propagation，再進一步處理 spatial misalignment，最後利用 object-level information 引導 dense feature fusion。由此可見，Asynchronous Feature Alignment 可進一步由 feature-level compensation 發展至 object-centric temporal alignment，透過更明確的物體 correspondence 與 motion information，降低 asynchronous information 對 cooperative perception 的影響。

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